Nischal A

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RESEARCH INTERESTS

Artificial Intelligence, Machine and Deep Learning, Natural Language Processing, Multi-modal Multimedia Analysis, Explainability and Robustness

EDUCATION BACKGROUND

Indian Institute of Technology Patna **July.2018-May.2022**
Bachelor of Technology (B.Tech), Computer Science and Engineering *CPI: 9.20/10*

PUBLICATIONS

* denotes equal contribution

- **Nischal Ashok***, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"A Multi-task Learning Approach for Fake News Detection: Novelty, Emotion, and Sentiment Lend a Helping Hand"**, International Joint Conference on Neural Networks, **IJCNN-2021**, (CORE-A) [**Accepted**]
- **Nischal Ashok***, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"Misinformation detection using multitask learning with mutual learning for novelty detection and emotion recognition"**, Journal of Information Processing and Management, **IPM**, (CORE-A) (IF:4.787) [Paper] [Code]
- **Nischal Ashok**, Jan Philip Wahle, Terry Ruas, Norman Meuschke, Tirthankar Ghosal and Bela Gipp, **"Empirical Studies on Health Misinformation Detection"**, (CORE-A*) [under review]
- Rina Kumari, **Nischal Ashok**, Tirthankar Ghoshal and Asif Ekbal, **"Multi-Modal Dataset for Fake News Detection"**, (CORE-A*) [under review]
- **Nischal Ashok***, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"What the fake? Probing Misinformation Detection Standing on the Shoulder of Novelty and Emotion"**, Journal of Information Processing and Management, **IPM**, (CORE-A) (IF:4.787) [under review]
- **Nischal Ashok**, Venkatesh Babu R, **"Exploring Adversarial Robustness in Deep Neural Networks"**, (Technical Report) Indian Academy of Sciences Research Portal [**Accepted**]

EXPERIENCE

Intern, IBM Research AI **IBM, Bangalore, India**
Research Managers: • ***Nitin Gupta*** • ***Hima Patel*** *May.2021-Present*

- **Project: Data Transformation Using Programming by Example**

Undergraduate Researcher, DKE Group **Universität Wuppertal, Germany**
Advisors: • ***Dr. Terry Ruas*** • ***Prof. Dr. Bela Gipp*** *Dec.2020-May.2021*

- **Project: Language Technologies for detecting health fake news**
- Performed and Analyzed Transfer Learning based NLP approaches on health-related fake-news datasets
- Work under review in a prestigious conference.

Undergraduate Researcher, AI-NLP-ML Lab **IIT-Patna, India**
Advisors: • ***Dr. Asif Ekbal*** • ***Prof. Pushpak Bhattacharya*** *July.2019-Present*

- **Present Project: Multi-modal Misinformation Detection**
- Probed uni-modal misinformation detection through novelty detection, emotion recognition and sentiment analysis.
- Proposed a novel multi-tasking deep neural network architecture for fake news detection giving SOTA results on 4 well known datasets.
- **Previous Work:** Document Level Novelty Detection and IIT-Patna Academic Alexa

AI Research Intern, Video Analytics Laboratory **Indian Institute of Science, India**
Advisor: • ***Prof. Venkatesh Babu*** *May.2020-Aug.2020*

- **Project: Unlabeled Data improves Adversarial Robustness**
- Investigated the role of unlabeled data for improving Adversarial Robustness in Deep Neural Networks using Cifar-10, Cifar-100, and TI-80M as the datasets.
- Experimented with the state-of-the-art adversarial deep learning algorithms based on semi-supervised learning.
- Probed the role of unlabeled data by proposing new loss functions, sourcing data from unrelated distributions, and generating it from DC-GANs and DE-GANs

AI Startup Winter Intern, TildeHat **Bangalore, India**
Manager: • ***Dr. Om Deshmukh*** *Dec.2019-Jan.2021*

- **Project: Unsupervised Data Retrieval**
- Designed a complete end-to-end module for useful information extraction from resumes.
- Implemented unsupervised algorithms and hierarchical classifier approaches for data-retrieval, conducting experiments on thousands of documents.

PROJECTS

IIT-Patna Academic Alexa **IIT Patna**
Guide: Prof. Pushpak Bhattacharya, Former Director IIT-P *Jan.2020-May.2020*

- A sentiment-aware intelligent information retrieval based Chatbot for the Academic Portal of IIT-Patna.
- Devised a novel and effective way of converting natural language data into SQL queries using Dependency Tree Parsers, POS tagging, and NER techniques
- Designed a custom chatbot pipeline which includes Multi-Channel CNN based intent classifier, sentiment analyzer, sentence similarity and feedback modules
- Designed a simple portable interface using wx-python GUI programming.

Reddit-Drug Abuse Detection **Data Analytics Lab, IIT Patna**
Guide: Dr. Joydeep Chandra *Jan.2020-May.2020*

- Investigated Drug Abuse on Social Media through Reddit Website by collecting and annotating data from r-opiates sub-reddit.
- Built a multi-head attention deep neural network for classifying comments as motivational or not. Further classification done on altruism, hope, good advice, bad advice, and universality.

Computer Science Academic Projects **IIT-Patna, India**
Guides: • Dr. Jimson Mathew • Dr. Samrat Mondal *May.2020-May.2021*

- **Two Pass Assembler and Emulator** - Computer Architecture CS321-322.
- **Smart Lock System** - Switching Theory CS225-226
- **Delivery Company Database System** - Database Laboratory Course CS355
- **HoneyPot based Authentication System** - Secure System Design CS392

POSITIONS OF RESPONSIBILITIES

Head-Machine Learning/ AI **IIT Patna, India**
NJACK, Computer Science Club *May.2020-Present*

- Introduced NJACK ML Workshop Series comprising sessions of Competitive ML and Research in ML
- Taught students about topics like Ensemble ML, Introduction to Deep Learning and others.

Departmental Lead **IIT Patna, India**
DSC IIT Patna, Google Developers *Aug.2019-May.2020*

- Associate Facilitator of Explore-ML Workshop IIT-P powered by Google-AI
- Promoted and participated in projects like Drone-Eye. Introduced ML to freshmen students.

Head-Industrial Relations **IIT Patna, India**
Entrepreneurship Cell *Aug.2019-May.2021*

- Lead the IR division managing a team of 40 volunteers and 20 sub-coordinators.
- Introduced Entrepreneur-101 series and organized more than 4 sessions on entrepreneurship.

AWARDS and HONORS

- **Conference Grant Scholarship** - ICWSM 2021-AAAI
- **Summer Research Fellow** (FASTSF'2020) - Indian Academy of Sciences [India's foremost Science Academy] for pursuing research internship in Artificial Intelligence at IISc, Bangalore.
- **IIT-Patna Merit List** - Among top 10 CPI holder in the IIT-Patna Computer Science class of 2022.
- **Fellowship (KVPY'2016)** - Kishore Vigyanik Protsahan Yojna - Government of India, for pursuing a career in research in pure sciences. Stood among top 0.2% among 200,000 students.
- **Prathiba Ratna Award** - Karnataka State (an Indian state) Education Development Authority for standing 2nd for the state in the 10th standard ICSE Board Exam 2016 among 150,000 students.

SKILLS

- **Programming Languages:** • C/C++ • Python • Java • Bash • Verilog • Assembly
- **Deep Learning Tools:** • Tensorflow • Keras • PyTorch • HuggingFace
- **Other tools:** • SQL • Docker • LaTeX • Git/GitHub
- **Operating Systems:** • Linux • Windows

SERVICE

- Mentor - Institute Student Mentorship Program (IIT-P) (2020-Present)
- Mentor - NJACK Winter of Code Open Source Programme'2019 - [ML Project Link]
- Social Worker - National Service Scheme IIT-P
- Teaching [Link] and writing blogs [Link] on ML and AI

CONFERENCES and TALKS

- Conferences - Attendee - ECIR'2021, ICON'2020, SPCOM'2020
- Seminars - Attendee - Stanford NLP (Spring'21), AMII Talks, AI for Social Impact - PSU